Name:_____ Period:_____

UNIT 2:

SYSTEMS OF MATTER AND ENERGY

ANCHOR CHARTS

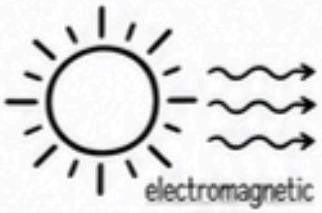


Energy in Earth's Systems and Heat Transfer

1 Earth's Two Energy Sources

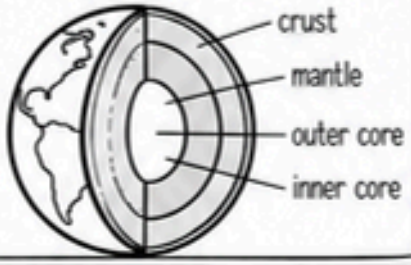
The Sun (external)

- Reaches Earth as electromagnetic energy.
- It drives weather, ocean currents, and the water cycle.



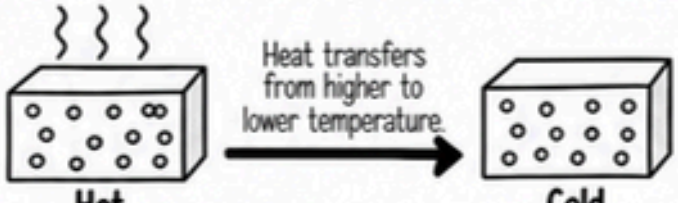
Earth's interior (internal)

- Heat comes mainly from radioactive decay of unstable elements.
- It drives mantle convection, plate movement, volcanoes, and geysers.



2 Heat vs. Temperature

- **Temperature** is the average kinetic energy of particles.
- **Heat** is the transfer of thermal energy from higher to lower temperature.
- Heat keeps flowing until both are equal.



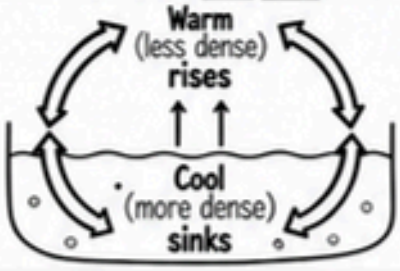
3 Three Methods of Heat Transfer

Method	How it moves	Needs matter?	Example
Conduction	Direct contact, particle to particle.	**Yes** (best in solids)	Hot ground warms a boardwalk.
Convection	Fluid moves because of density differences.	**Yes** (liquids, gases)	Hot water rising in a hot spring.
Radiation	Electromagnetic waves.	**No**	Sunlight warming snow.

Key Point:
Radiation is the only method of heat transfer that can cross space from the Sun to Earth.

4 Convection Currents and Density

- Warm fluid expands, becomes less dense, and rises.
- Cool fluid becomes more dense and sinks.

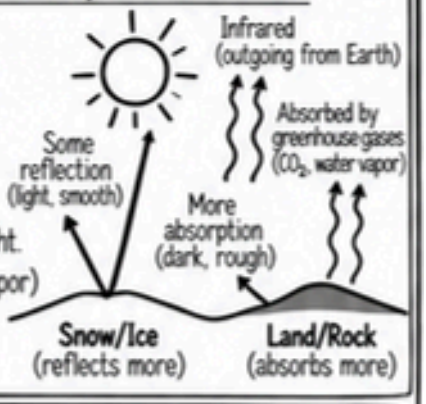


Where it happens:

- Atmosphere (wind)
- Oceans (currents)
- Mantle (asthenosphere, which moves the plates)

5 Absorption, Reflection, and Infrared

- Dark, rough surfaces absorb more energy.
- Light, smooth surfaces like snow and ice reflect more energy.
- Earth's surface gives off energy as infrared radiation, which has a longer wavelength than visible light.
- Greenhouse gases (CO_2, water vapor) absorb infrared and keep the atmosphere warmer.



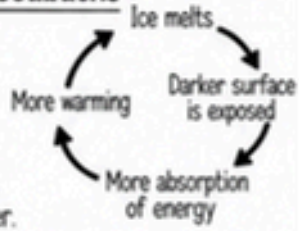
6 Specific Heat

- Specific heat is the energy needed to raise 1 g of a substance by 1°C.
- High specific heat means a substance heats up slowly and cools slowly, like water.
- Low specific heat means it heats up quickly and cools quickly, like land and sand.



7 Energy Balance and Feedbacks

- Energy in = energy out, so the temperature stays steady.
- More in than out → Earth warms.
- More out than in → Earth cools.
- A feedback is a change that makes the original change bigger or smaller.



Part A: Fill in the Blank

1. _____ is the average kinetic energy of particles.
2. Heat inside Earth comes mainly from the _____ of unstable elements.
3. Earth's surface gives off energy as _____ radiation, which has a longer wavelength than visible light.
4. _____ is the energy needed to raise 1 g of a substance by 1°C.
5. A _____ is a change that makes the original change bigger or smaller.

Part B: Matching

Match each term to its example.

Term	Answer	Example
1. Conduction	_____	A. Sunlight warming snow
2. Convection	_____	B. CO_2 and water vapor keeping the atmosphere warmer
3. Radiation	_____	C. Hot ground warming a boardwalk
4. Greenhouse gases	_____	D. Water heating up slowly and cooling slowly
5. High specific heat	_____	E. Hot water rising in a hot spring

Part C: True / False

1. _____ Radiation is the only method of heat transfer that can cross space from the Sun to Earth.
2. _____ Light, smooth surfaces like snow and ice absorb more energy than dark, rough surfaces.
3. _____ Warm fluid expands, becomes less dense, and rises.
4. _____ When more energy comes in than goes out, Earth cools.

Part D: Short Answer / Application

Use what you know, along with the anchor chart, to answer the following.

1. A metal spoon sits in a pot of soup on the stove. The spoon's handle gets hot, and the soup circulates as it heats. Name the method of heat transfer for the spoon and for the soup, and explain how the energy moves in each.

   Spoon1: _____ Soup: _____

   _____

   _____

   _____

2. On a sunny afternoon at the beach, the sand feels hot, but the lake water still feels cool. Use specific heat to explain why.

   _____

   _____

3. Arctic sea ice is melting. Use the feedback loop on the anchor chart to explain how melting ice can lead to even more warming.

   _____

   _____

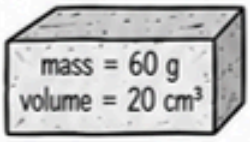
DENSITY, PHASE CHANGES, AND MATERIAL BEHAVIOR

1 Density Basics

- **Density** = **mass** ÷ **volume** (g/cm³).
- This formula is not on the reference tables, so you need to know it.
- Size does not matter. A chip and a boulder of the same rock have the same density.
- Less dense floats or rises. More dense sinks.

$$D = \frac{m}{V}$$

D = density (g/cm³)
m = mass (g)
V = volume (cm³)

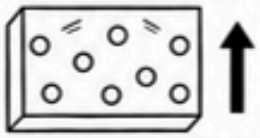


$$D = \frac{60\ g}{20\ cm^3} = 3.0\ g/cm^3$$

2 Temperature and Density

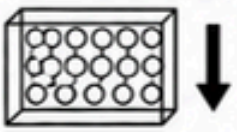
- **Heating**: particles spread out, the material expands, and its density decreases, so it rises.
- **Cooling**: particles move closer, the material contracts, and its density increases, so it sinks.
- Hot magma is less dense than the solid rock around it, so it rises toward the surface.

Heated (less dense)



- Particles spread out
- Expands
- Density decreases

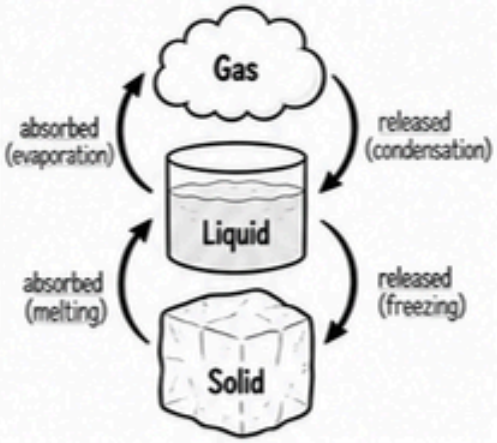
Cooled (more dense)



- Particles move closer
- Contracts
- Density increases

3 Phase Changes and Energy

Phase change	Change of state	Energy	Example
Melting	solid → liquid	absorbed	Rock melts into magma
Freezing (crystallization)	liquid → solid	released	Lava hardens into basalt
Evaporation	liquid → gas	absorbed	Rain on hot ground turns to steam
Condensation	gas → liquid	released	Steam cools into mist



- **Absorbed** means the particles speed up and break apart.
- **Released** means the particles slow down and lock together.
- **Crystallization** is when mineral crystals form as magma or lava freezes.

4 Water Is Unusual

- Most materials are more dense as solids. Water is the exception.
- Water expands about 9% when it freezes, so ice is less dense and floats.
- This expansion breaks rock through frost wedging, which comes back in a later unit.



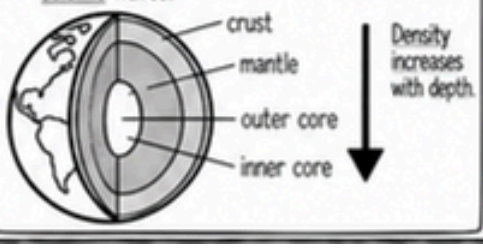
Ice is less dense than liquid water, so it floats.



Water expands when it freezes, which can break rock (frost wedging).

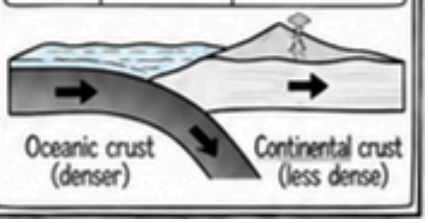
5 Density and Earth's Layers

- Early Earth was hot enough to melt. The densest materials (iron and nickel) sank to the center, and less dense materials rose.
- Layers from the outside in: crust → mantle → outer core → inner core.
- Density increases with depth.
- The mantle is solid but flows slowly (a few cm per year) because of high temperature and pressure.
- Scientists learn about the layers from seismic waves.



6 Oceanic vs. Continental Crust

	Oceanic crust	Continental crust
Main rock	Basalt (dark)	Granite (light-colored)
Density	about 3.0 g/cm³	about 2.7 g/cm³
When they collide	Sinks beneath	Stays on top



DENSITY, PHASE CHANGES, AND MATERIAL BEHAVIOR

Part A: Fill in the Blank

1. Density equals _____ divided by volume.
2. When a material is heated, its particles spread out, it expands, and its density _____.
3. _____ is when mineral crystals form as magma or lava freezes.
4. Water expands about _____ when it freezes, so ice is less dense and floats.
5. Scientists learn about Earth's layers from _____.

Part B: Matching

Match each term to its example.

Term	Answer	Example
1. Melting	_____	A. Steam cools into mist
2. Freezing (crystallization)	_____	B. Water freezes in a crack, expands, and breaks the rock
3. Evaporation	_____	C. Rock melts into magma
4. Condensation	_____	D. Rain on hot ground turns to steam
5. Frost wedging	_____	E. Lava hardens into basalt

Part C: True / False

1. _____ The density formula is on the reference tables.
2. _____ A small chip of granite has the same density as a large boulder of granite.
3. _____ Density decreases with depth inside Earth.
4. _____ When oceanic crust and continental crust collide, the oceanic crust sinks beneath.

Part D: Short Answer / Application

Directions: Answer in complete sentences. Use vocabulary from the anchor chart in your answer.

1. A rock sample has a mass of 90 g and a volume of 30 cm^3. Calculate its density and show your work. Then explain whether a small chip broken off this rock would have a different density.

   Density: _____ g/cm^3

   _____

   _____

2. A glass bottle completely filled with water is left in a freezer overnight. In the morning, the bottle is cracked. Explain why, using what makes water unusual.

   _____

   _____

3. A plate made of rock with a density of about 3.0 g/cm^3 collides with a plate made of rock with a density of about 2.7 g/cm^3. Predict which plate will sink, and identify the type of crust and main rock in each plate. Explain your reasoning.

   _____

   _____

MINERALS

1 What Is a Mineral? (5 Requirements)

- ☐ **Naturally occurring:** forms in nature, not in a factory or lab.
- ☐ **Inorganic:** not made by living things or from their remains.
- ☐ **Solid:** has a definite shape and volume.
- ☐ **Crystal structure:** atoms in an orderly, repeating pattern.
- ☐ **Definite chemical composition:** always the same elements in the same proportions (calcite is always $CaCO_3$).

If a substance is missing even one requirement, it is not a mineral.

Minerals are the building blocks of rocks.
Granite, for example, is made mostly of quartz and feldspar.

Mineral or Not?



Calcite ($CaCO_3$)
✓ **Mineral**



Factory glass (made in a factory)
✗ Not a mineral



Coal (from plant remains)
✗ Not a mineral

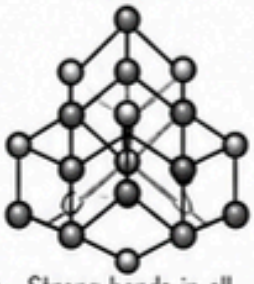


Obsidian (formed from rapid cooling)
✗ Not a mineral

2 Structure Determines Properties

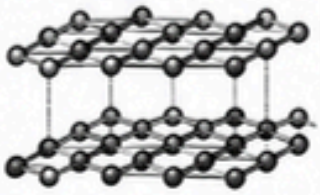
- How a mineral's atoms are arranged and bonded controls its crystal shape, its hardness, and how it breaks.
- Diamond and graphite are both pure carbon, but their atomic structure is different.

Diamond



- Strong bonds in all directions.
- Very hard (hardest natural mineral).
- Each carbon atom is bonded to 4 other carbon atoms.

Graphite



- Flat sheets with weak bonds between layers.
- Layers slide apart easily.
- Soft (used as pencil "lead").
- Each carbon atom is bonded to 3 other carbon atoms in a sheet.

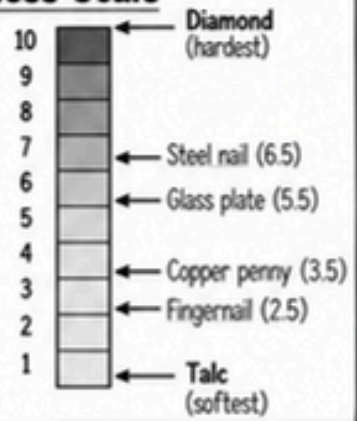
3 Identifying Minerals by Their Properties

Property	What it is	Example
Color	Easy to see, but often unreliable.	Quartz can be clear, white, pink, or purple.
Streak	Color of the powder on a streak plate.	Hematite leaves a reddish-brown streak.
Luster	How the surface reflects light (metallic or nonmetallic).	Pyrite is metallic, calcite is glassy.
Hardness	Resistance to scratching.	Gypsum is scratched by a fingernail.
Cleavage	Breaks along smooth, flat surfaces (weak bonds).	Halite breaks into cubes.
Fracture	Breaks along rough or curved surfaces.	Quartz breaks with curved, shell-like surfaces.
Crystal shape	Controlled by the atomic arrangement.	Halite grows as cubes, quartz as six-sided columns.
Density	Mass ÷ volume, the same for any sample size.	Pyrite (about 5.0 g/cm³) vs. quartz (about 2.7 g/cm³).
Other tests	Special properties.	Calcite fizzes in acid, magnetite is magnetic.

★ **Key Point:** **Streak is more reliable than color** because impurities can change a mineral's color.

4 Mohs Hardness Scale

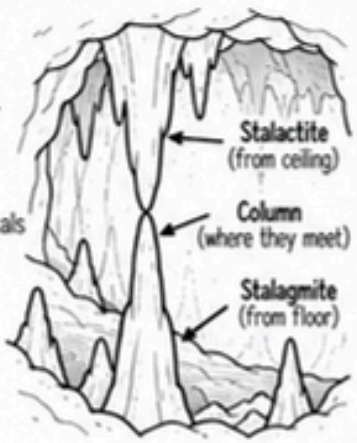
- The scale runs from 1 (talc, softest) to 10 (diamond, hardest).
- A mineral or tool scratches anything with a lower hardness number.
- Common test tools:
  - Fingernail 2.5
  - Copper penny 3.5
  - Glass plate 5.5
  - Steel nail 6.5



5 How Minerals Form

From magma or lava (crystallization):
- Atoms lock into crystals as the melt cools.
- Slow cooling underground grows larger crystals.

From water (precipitation):
- Dissolved minerals come out as solid crystals when the water evaporates or its conditions change.
- Carlsbad Caverns: calcite drips build stalactites (down from the ceiling), stalagmites (up from the floor), and columns (where the two meet).
- White Sands: gypsum crystals are left behind when the water evaporates.



Part A: Fill in the Blank

1. A mineral must be _____, which means it forms in nature, not in a factory or lab.
2. _____ is the color of a mineral's powder on a streak plate.
3. Diamond and graphite are both pure _____, but their atomic structure is different.
4. The Mohs Hardness Scale runs from 1, _____ (softest), to 10, diamond (hardest).
5. Slow cooling underground grows _____ crystals.

Part B: Matching

Match each property to its example.

Property	Answer	Example
1. Streak	_____	A. Quartz breaks with curved, shell-like surfaces
2. Luster	_____	B. Hematite leaves a reddish-brown streak
3. Hardness	_____	C. Calcite fizzes in acid, and magnetite is magnetic
4. Cleavage	_____	D. Pyrite is metallic, and calcite is glassy
5. Fracture	_____	E. Halite breaks into cubes
6. Other tests	_____	F. Gypsum is scratched by a fingernail

Part C: True / False

1. _____ Coal is a mineral because it forms naturally.
2. _____ Calcite always has the same chemical composition: $CaCO_3$.
3. _____ A glass plate can scratch a mineral with a hardness of 4.
4. _____ Stalactites build up from the cave floor, and stalagmites hang down from the ceiling.
5. _____ Color is the most reliable property for identifying a mineral.

Part D: Short Answer / Application

Use what you know, along with the anchor chart, to answer the following.

1. Diamond and graphite are both made of pure carbon. Explain why diamond is very hard, but graphite is soft enough to use in pencils.

_____

_____

2. *A geologist finds two samples. Sample 1 has large crystals and formed deep underground from magma. Sample 2 is gypsum crystals left behind in a dry lake bed. Identify how each sample formed and explain.*

*Sample 1:*_____ *Sample 2:* _____

_____

_____

IGNEOUS ROCKS AND MAGMA

1 How Magma Forms

- **Igneous rock** forms when melted rock cools and hardens.
- **Magma** is below the surface, and **lava** is at the surface.
- Rock melts in three ways:

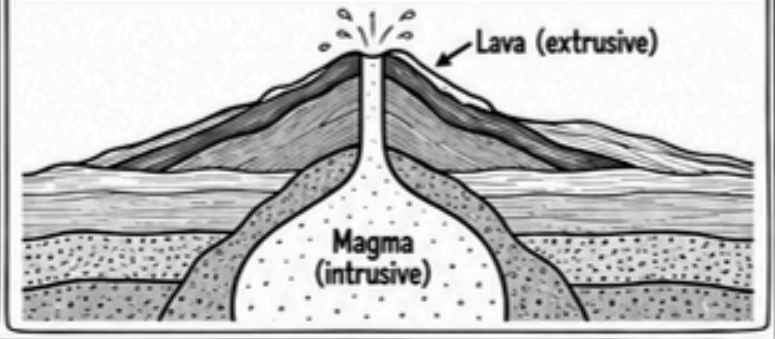
Increasing temperature	Decreasing pressure	Adding water
		
The rock gets hot enough to partly melt.	Hot mantle rock rises, the pressure drops, and it melts without getting hotter (Hawai'i).	Water lowers the melting temperature of rock, the same way salt lowers the melting point of ice (subduction, Yosemite).

- **Magma** is **less dense** than the rock around it, so it rises.

2 Cooling Rate and Crystal Size

- **Slow cooling** gives large crystals.
- **Fast cooling** gives small crystals or none.

Type	Where it cools	Speed	Texture	Examples
Intrusive	Underground	Slow	Coarse-grained	Granite, gabbro, pegmatite (very coarse)
Extrusive	At the surface	Fast	Fine-grained	Basalt, rhyolite



3 Special Textures

Glassy

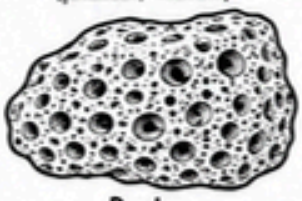
- Cooled so fast that no crystals formed (obsidian).



Obsidian

Vesicular

- Holes left by trapped gas bubbles (pumice, scoria).



Pumice

Porphyritic

- Large crystals surrounded by tiny ones, from two cooling rates (slow underground, then fast at the surface).



Porphyritic Andesite

4 Composition: Felsic vs. Mafic

	Felsic	Mafic
Rich in	Silica	Iron and magnesium
Minerals	Quartz, potassium feldspar	Pyroxene, olivine
Color	Light	Dark
Density	Less dense	More dense
Where found	Continental crust	Oceanic crust
Rock pairs (intrusive / extrusive)	Granite / rhyolite	Gabbro / basalt

- **Intermediate rocks** (diorite, andesite) fall in between.
- **Ultramafic rocks** (peridotite) come from the mantle.

★ **Key Point:** The same minerals with different cooling rates give different rocks (granite and rhyolite).

5 Bowen's Reaction Series

- Bowen's Reaction Series is the order minerals crystallize as magma cools from about 1,400°C to 650°C.
- Plagioclase feldspar crystallizes at the same time, starting calcium-rich and becoming sodium-rich as the magma cools.



★ **Key Point:** Early, high-temperature minerals make mafic and ultramafic rocks. Late, low-temperature minerals make felsic rocks.

IGNEOUS ROCKS AND MAGMA

Part A: Fill in the Blank

1. Magma is below the surface, and _____ is at the surface.
2. Slow cooling gives _____ crystals.
3. A _____ texture has holes left by trapped gas bubbles.
4. Felsic rocks are rich in _____, while mafic rocks are rich in iron and magnesium.
5. _____ is the order minerals crystallize as magma cools from about 1,400°C to 650°C.

Part B: Matching

Match each term to its example.

Term	Answer	Example
1. Decreasing pressure	_____	A. Peridotite, which comes from the mantle
2. Adding water	_____	B. Obsidian, which cooled so fast that no crystals formed
3. Glassy texture	_____	C. Rock melts at a lower temperature at a subduction zone (Yosemite)
4. Vesicular texture	_____	D. Andesite with large crystals surrounded by tiny ones
5. Porphyritic texture	_____	E. Hot mantle rock rises and melts without getting hotter (Hawai'i)
6. Ultramafic rock	_____	F. Pumice and scoria

Part C: True / False

1. _____ Granite is an extrusive rock that cooled quickly at the surface.
2. _____ All igneous rocks have visible crystals.
3. _____ Gabbro and basalt are both mafic, but basalt cooled faster.
4. _____ Felsic rocks are light-colored, less dense, and found in continental crust.
5. _____ Olivine and pyroxene crystallize late, at low temperatures, as magma cools.

Part D: Short Answer / Application

Use what you know, along with the anchor chart, to answer the following.

1. Rock A and Rock B are made of the same minerals. Rock A has large crystals you can easily see. Rock B has crystals too small to see. Identify where each rock cooled and explain how you know.

   Rock A: _____ Rock B: _____

   _____

   _____

2. A rock sample is dark, dense, and contains olivine and pyroxene. Classify it as felsic or mafic. Use Bowen's Reaction Series to explain when its minerals crystallized.

   Classification:_____

   _____

   _____

3. Under Hawai'i, hot mantle rock melts even though it does not get any hotter. Explain how this rock melts and why the magma then rises toward the surface.

   _____

   _____

SEDIMENTARY ROCKS

1 From Sediment to Rock (5 Steps)

- Sedimentary rocks form at or near Earth's surface.

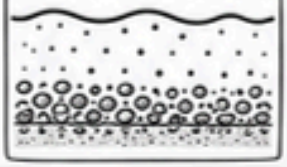
1. Weathering
Rock is broken into smaller pieces called sediment.



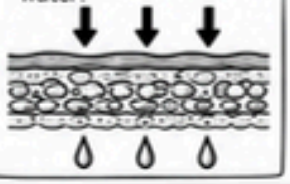
2. Erosion
Water, wind, or ice carries the sediment away.



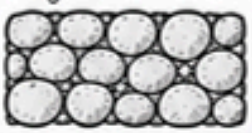
3. Deposition
The water or wind slows down, loses energy, and drops the sediment.



4. Compaction
The weight of the layers above squeezes the sediment together and presses out water.



5. Cementation
Minerals dissolved in the water (calcite, quartz) crystallize between the grains and glue them together.



2 Three Kinds of Sedimentary Rock

Kind	How it forms	Rocks	Example Image
Clastic	Fragments of other rocks, named by grain size.	Conglomerate, sandstone, siltstone, shale	
Chemical	Minerals crystallize out of water, often as it evaporates (evaporites).	Rock salt (halite), rock gypsum (White Sands)	
Organic	Remains of living things.	Limestone (shells, calcite), coal (plants)	

3 Clastic Grain Size (largest to smallest)

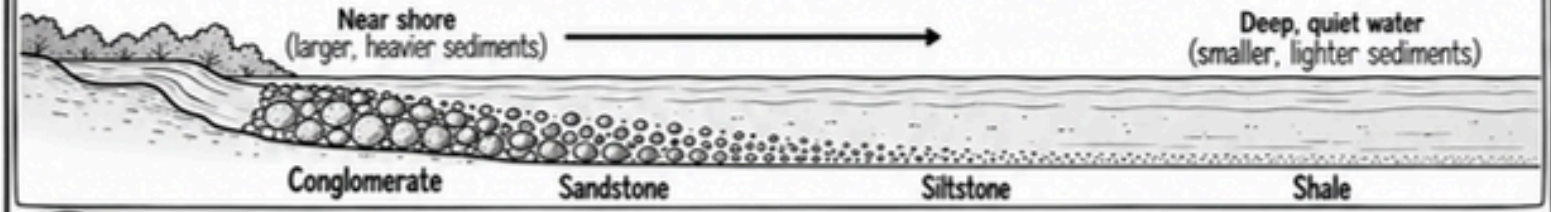
Pebbles/Gravel (conglomerate) → **Sand** (sandstone) → **Silt** (siltstone) → **Clay** (shale)



Largest grain size → Smallest grain size

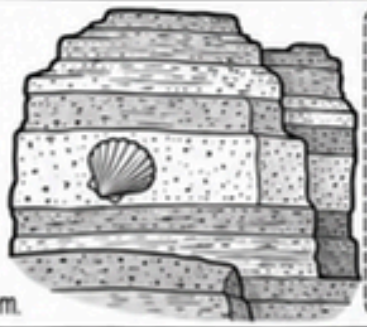
4 Horizontal Sorting

- When a river enters a lake or ocean, the water slows down and loses energy.
- The largest, heaviest sediments settle first, close to shore.
- The smallest sediments (clay) settle last, in deep, calm water.
- This is why conglomerate forms near shore and shale forms in deep, quiet water.



5 Layers and Fossils

- Sediment settles in flat layers called **strata**.
- Each new layer goes on top of older ones (the Badlands stripes).
- Strata are not the same as **bands**. Bands in gneiss are minerals that separated during metamorphism.
- Most fossils are found in sedimentary rock because sediment buries remains gently without destroying them.



Strata (in sedimentary rock)



Flat layers of sediment on top of each other.

Bands (in gneiss, a metamorphic rock)



Mineral bands formed during metamorphism (not layers of sediment).

SEDIMENTARY ROCKS

Part A: Fill in the Blank

1. Sedimentary rocks form at or near Earth's _____.
2. During _____, the weight of the layers above squeezes the sediment together.
3. _____ sedimentary rocks form when minerals crystallize out of water.
4. Sediment settles in flat layers called _____.
5. The smallest sediments, _____, settle last in deep, calm water.

Part B: Matching

Match each term to its example.

Term	Answer	Example
1. Weathering	_____	A. Water slows down, loses energy, and drops the sediment
2. Erosion	_____	B. Conglomerate, sandstone, siltstone, and shale
3. Deposition	_____	C. Rock is broken into smaller pieces called sediment
4. Cementation	_____	D. Limestone made of shells, and coal made from plants
5. Organic rock	_____	E. Water, wind, or ice carries sediment away
6. Clastic rock	_____	F. Calcite or quartz crystallizes between grains and glues them

Part C: True / False

1. _____ Sediment buries remains gently, which is why most fossils are found in sedimentary rock.
2. _____ Rock salt and rock gypsum are clastic sedimentary rocks.
3. _____ The grains in sandstone are larger than the grains in conglomerate.
4. _____ When a river enters a lake, the largest, heaviest sediments settle close to shore.
5. _____ The bands in gneiss are strata, just like the stripes in the Badlands.

Part D: Short Answer / Application

Use what you know, along with the anchor chart, to answer the following.

1. A river flows into a calm lake. Conglomerate forms near the shore, and shale forms in deep water. Explain why.

_____

_____

1. A rock is made of shells that were cemented together. Classify the kind of sedimentary rock, name the rock, and explain.

Kind: _____ Rock: _____

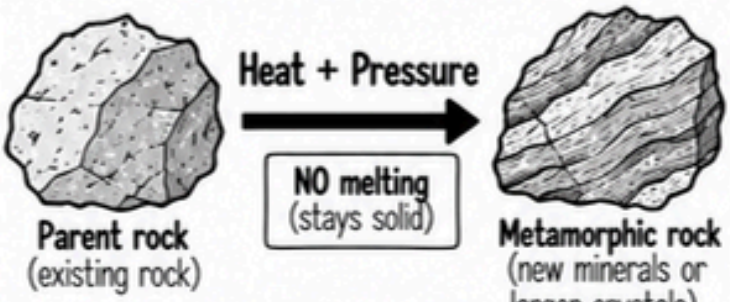
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METAMORPHIC ROCKS

1 Heat and Pressure Without Melting

- A **metamorphic rock** forms when an existing rock (the **parent rock**) is changed by **heat and pressure**.
- The rock **stays solid** the whole time. If it melted, it would become magma and later an igneous rock.
- **Recrystallization:** atoms rearrange into new or larger crystals while the rock stays solid.



2 Two Types of Metamorphism

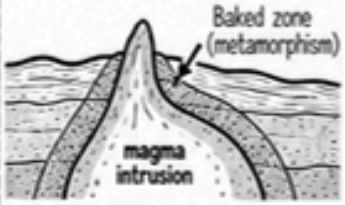
	Regional	Contact
Area	Large	Small
Cause	Deep burial and squeezing as plates collide and build mountains	Rock touches hot magma and is "baked"
Example	Great Smoky Mountains	Rock next to a magma intrusion

Regional Metamorphism



Deep burial and squeezing as plates collide and build mountains (large area).

Contact Metamorphism

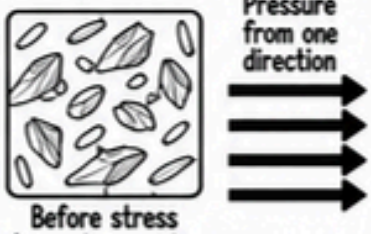


Rock touches hot magma and is "baked" (small area).

3 Foliated vs. Nonfoliated

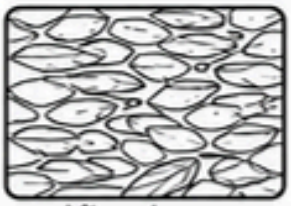
Foliated

- Pressure from one direction squeezes flat minerals (like mica) until they line up in parallel sheets.
- These rocks often split into flat pieces.



Nonfoliated

- Made mostly of one mineral that does not line up.
- Examples:
  - Sandstone → quartzite
  - Limestone → marble

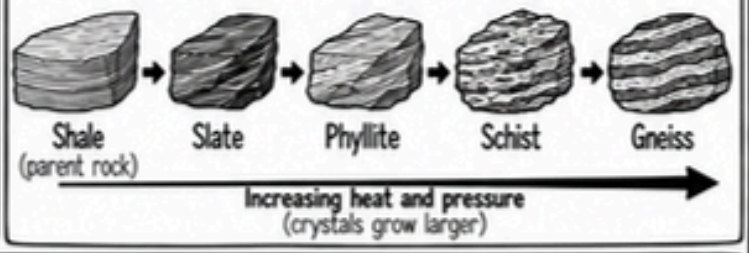


After stress
(minerals lined up
in parallel sheets)

4 Shale's Path (Increasing Heat and Pressure)

Rock	Heat and pressure	What you observe
Slate	Low	Dull, fine-grained, splits into thin, flat sheets.
Phyllite	Low to medium	Silky shine from tiny mica crystals.
Schist	Medium to high	Sparkly mica flakes large enough to see.
Gneiss	Highest	Light and dark bands.

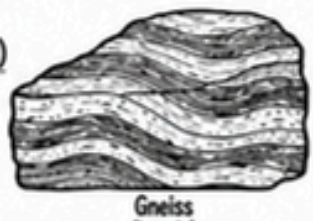
Shale → Slate → Phyllite → Schist → Gneiss



5 Bands vs. Layers

Bands (metamorphic rock - gneiss)

- Minerals that separated during metamorphism.
- Found **only** at the highest heat and pressure.



Gneiss
(bands)

Layers/strata (sedimentary rock)

- Sediment deposited one layer on top of another.
- Each layer is a different time in Earth's history.



Sandstone
(layers/strata)

Smokies connection: The rocks are mostly slate and phyllite, so they split into sheets but do not show bands.

METAMORPHIC ROCKS

Part A: Fill in the Blank

1. The existing rock that a metamorphic rock forms from is called the _____.
2. _____ is when atoms rearrange into new or larger crystals while the rock stays solid.
3. _____ metamorphism happens when rock touches hot magma and is "baked."
4. In foliated rocks, pressure from one direction squeezes flat minerals like _____ until they line up in parallel sheets.
5. Shale's path is shale → slate → phyllite → _____ → gneiss.

Part B: Matching

Match each rock to its example.

Rock	Answer	Example
1. Slate	_____	A. Sparkly mica flakes large enough to see
2. Phyllite	_____	B. Forms from sandstone
3. Schist	_____	C. Light and dark bands
4. Gneiss	_____	D. Dull, fine-grained, and splits into thin, flat sheets
5. Marble	_____	E. Forms from limestone
6. Quartzite	_____	F. Silky shine from tiny mica crystals

Part C: True / False

1. _____ If a rock melts during metamorphism, it becomes a metamorphic rock with larger crystals.
2. _____ Regional metamorphism affects a large area as plates collide and build mountains.
3. _____ Marble is foliated because its minerals line up in parallel sheets.
4. _____ Slate forms at the highest heat and pressure along shale's path.
5. _____ Most rocks in the Great Smoky Mountains split into sheets but do not show bands.

Part D: Short Answer / Application

Use what you know, along with the anchor chart, to answer the following.

1. A student finds two striped rocks. Rock A has light and dark bands. Rock B has flat layers of sand. Identify each rock type and explain how the stripes formed in each.

   Rock A: _____ Rock B: _____

   _____

   _____

2. A layer of limestone sits right next to a magma intrusion. Predict the rock that forms in the baked zone, and name the type of metamorphism. Explain.

   Rock: _____ Type: _____

   _____

   _____

3. The Great Smoky Mountains are mostly slate and phyllite. What does this tell you about the heat and pressure these rocks experienced? Explain why they do not show bands.

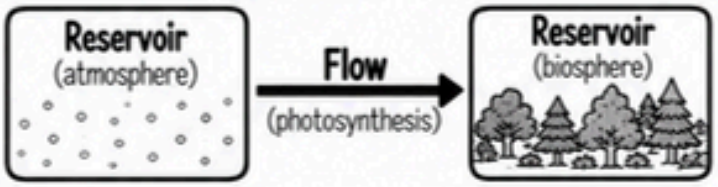
   _____

   _____

CYCLES AND SYSTEM MODELS

1 Parts of a System Model

- A **system** is a group of parts that work together.
- A system model has three parts:
  - **Reservoirs**: places where matter is stored (atmosphere, ocean, living things, rocks)
  - **Flows**: processes that move matter between reservoirs (photosynthesis, evaporation)
  - **Inputs and outputs**: matter or energy entering or leaving the system.



2 Matter vs. Energy in Earth's System

	Matter	Energy
Type of system	Almost closed	Open
What happens	Conserved, moves from reservoir to reservoir	Enters from the Sun and Earth's interior
Role	Is cycled	Drives the flows

Examples of matter cycles:

- Rock cycle
- Water cycle
- Carbon cycle

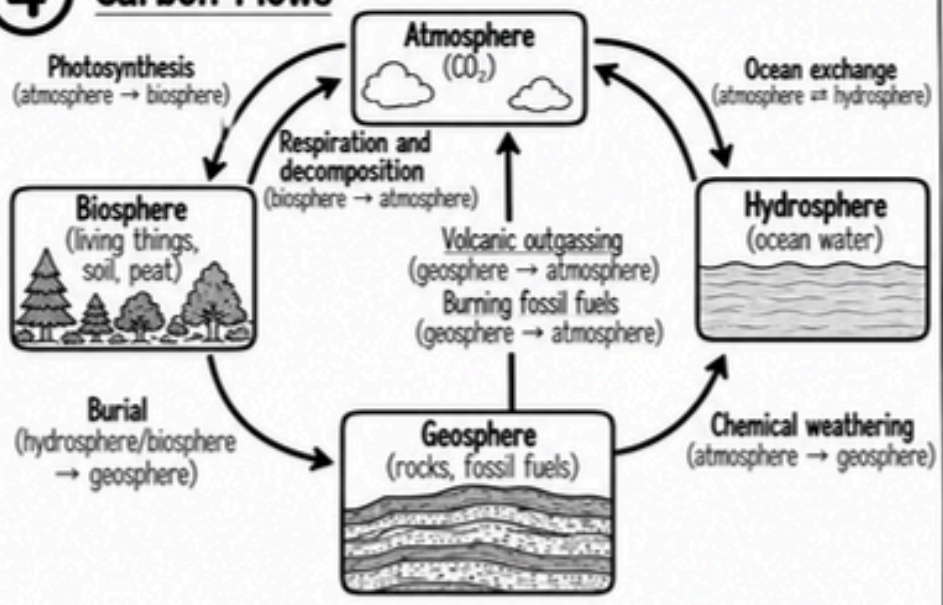
3 Carbon Reservoirs

(measured in gigatons of carbon, GtC, where 1 GtC is one billion metric tons)

Reservoir	Where carbon is stored	About how much	How long it stays
Atmosphere	CO_2 gas in the air	900 GtC	About 5 years
Biosphere	Living things, dead plants, soil, peat	2,100 GtC	10 to 100 years
Hydrosphere	Dissolved in ocean water	38,000 GtC	Hundreds to about 1,000 years
Geosphere	Limestone, fossil fuels	60,000,000+ GtC (largest)	100+ million years

★ **Pattern**: Larger reservoirs usually hold carbon for a longer time.

4 Carbon Flows



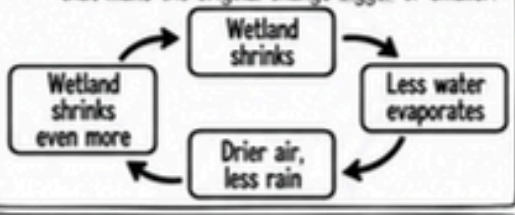
5 Fast vs. Slow Carbon Cycle

	Fast	Slow
Time	Years to centuries	Millions of years
Size of yearly flow	Huge	Small
Examples	Photosynthesis, respiration, ocean exchange	Chemical weathering, burial, volcanoes

★ **Key Point**: Slow flows are small each year, but they control the largest reservoir over millions of years.

6 Feedbacks

A **feedback** is a change that causes effects that make the original change bigger or smaller.



7 Everglades Connection



- Waterlogged soil has little oxygen, so dead plants decay slowly. The plant material builds up as peat, which stores carbon (a "carbon bank").
- Draining exposes the peat to air, so decomposers release the carbon as CO_2 (biosphere → atmosphere).
- Peat buried for millions of years can become coal.



Part A: Fill in the Blank

1. _____ are places where matter is stored, such as the atmosphere and ocean.
2. For matter, Earth is an almost _____ system.
3. The largest carbon reservoir is the _____.
4. During _____, carbon moves from the atmosphere to the biosphere.
5. In waterlogged soil, dead plants decay slowly and build up as _____.

Part B: Matching

Match each term to its definition.

Term	Answer	Example
1. Respiration and decomposition	_____	A. Atmosphere → geosphere
2. Volcanic outgassing	_____	B. A wetland shrinks, so it shrinks even more
3. Chemical weathering	_____	C. Biosphere → atmosphere
4. Burial	_____	D. Atmosphere ↔ hydrosphere (both ways)
5. Ocean exchange	_____	E. Geosphere → atmosphere through volcanoes
6. Feedback	_____	F. Hydrosphere or biosphere → geosphere

Part C: True / False

1. _____ Carbon stays in the atmosphere longer than it stays in the geosphere.
2. _____ Energy enters Earth's system from the Sun and Earth's interior and drives the flows.
3. _____ Slow carbon flows are small each year, but they control the largest reservoir.
4. _____ Photosynthesis and respiration are examples of the slow carbon cycle.
5. _____ Peat buried for millions of years can become limestone.

Part D: Short Answer / Application

Use what you know, along with the anchor chart, to answer the following.

1. The hydrosphere holds about 38,000 GtC, and the atmosphere holds about 900 GtC. Which holds carbon longer? Explain using the pattern on the anchor chart.

   Reservoir: _____

   _____

   _____

2. Part of the Everglades is drained for farmland. Identify the reservoir the carbon leaves and the one it enters. Explain how it moves.

   Leaves: _____ Enters: _____

   _____

   _____

3. Explain how matter and energy behave differently in Earth's system. Use carbon as your example of matter.

   _____

   _____